

PERFORMANCE TASK 3

File System Administration: Provisioning, Partitioning and Mounting Storage

IT 4112 — System Administration and Maintenance • Week 4 • Laboratory

Name:		Section:	
Date Performed:		Date Submitted:	
Duration:	2 hours 30 minutes	Total Points:	100

1 The Situation

ISLA VERDE TRADING CORP. — WORK ORDER #IVT-2026-041

Isla Verde Trading Corp. has 40 employees across Sales, Finance and Warehouse. Its single Ubuntu Server has been running on one 25 GB virtual disk since it was set up by a contractor who has since left. Nobody documented anything.

Last Tuesday the server started throwing "No space left on device" errors. The Finance team could not save month-end reports. Management approved the purchase of two new disks, and they have now been handed to you — still blank, still unpartitioned, still invisible to the operating system.

Your job as junior system administrator: bring both disks into service, give each department a properly formatted and mounted storage area, prove that files can be written to each one, then take the storage back offline cleanly at the end of your shift so the Data Center team can move the physical trays.

The two disks you have been given

Disk	Capacity	Partition Plan	File System	Intended Department
Storage 1	20 GB	1 partition — full disk	ext4	Company archive
Storage 2	30 GB	Partition 1 — 15 GB	ext4	Finance records
		Partition 2 — remaining ~15 GB	NTFS	Windows exchange folder

WHY NTFS ON A LINUX SERVER?

Finance still runs two Windows accounting workstations. Files placed on the NTFS partition can be read directly if that disk is ever moved to a Windows machine. This is a deliberate design decision, not a mistake — and one of the concept questions will ask you to defend it.

2 What You Should Be Able to Do Afterwards

By the end of this session you should be able to:

- Attach new virtual hard disks to an existing VirtualBox guest and confirm the operating system sees them.
- Read block device names correctly and explain what each part of a name such as `/dev/sdc2` means.
- Create partitions on a raw disk using `fdisk`, including a mixed layout on a single disk.
- Format partitions with the `ext4` and `NTFS` file systems and explain why you would pick one over the other.
- Create mount points, mount partitions, write data to them, and unmount them without corrupting anything.
- Document your work with evidence that another administrator could verify.

3 Before You Start

3.1 What you need

Item	Requirement
Host machine	Any machine running VirtualBox 7.x with at least 60 GB free on the physical drive
Virtual machine	Your existing Ubuntu Server guest from PT1 and PT2, powered off
Account	A user account with <code>sudo</code> privileges
Internet	Needed once, to install the <code>ntfs-3g</code> package
Software	Microsoft Word (or Google Docs / LibreOffice Writer exporting to <code>.docx</code>)

3.2 Take a snapshot first — this is not optional

You are about to run commands that permanently erase disks. A snapshot is your undo button. Take it while the VM is powered OFF.

```
VirtualBox Manager
# In the VirtualBox Manager window:
#   Right-click your VM -> Snapshots -> Take (camera icon)
#   Name it exactly: PT3-START
#   Description:     before adding storage 1 and storage 2
```

THE ONE RULE THAT MATTERS TODAY

Every destructive command in this task names a specific device. `/dev/sda` is your existing system disk. If you ever type `sda` where the instruction says `sdb` or `sdc`, you will destroy the operating system you are working on. Read the device name out loud before you press Enter.

3.3 How to spend your two and a half hours

Part	Activity	Time
Setup	Snapshot, power off, read through the whole task once	10 min

Part	Activity	Time
Part A	Attach the two virtual disks in VirtualBox	15 min
Part B	Boot and identify the new disks	10 min
Part C	Partition and format Storage 1	20 min
Part D	Partition and format Storage 2 (two partitions, two file systems)	35 min
Part E	Create mount points and mount all three partitions	20 min
Part F	Write one text file to each partition and verify	15 min
Part G	Unmount everything and confirm	10 min
Part H	Answer the concept questions	15 min
	TOTAL	2 h 30 min

4 How Your Evidence Must Look

Every part of this task asks for one or more screenshots. Fourteen screenshots are required in total. They are the only proof that you actually did the work, so they have to be readable and they have to be yours.

Rules for screenshots

- Capture the whole terminal window or the whole VirtualBox dialog — not a cropped fragment of one line.
- The command AND its output must both be visible in the same shot.
- Before your first screenshot, run `whoami` once so your username appears in the shell prompt. That prompt must be visible in every later screenshot.
- Do not retype output into the document. A pasted block of text is not a screenshot and earns zero.
- Insert images directly into the Word file, sized so the text is readable at 100% zoom. Do not attach them as separate files.
- Label each image with its code (SS-01, SS-02, ...) as a caption underneath.

Required screenshot list

Code	What it must show	Part
SS-01	VirtualBox Settings → Storage with both new disks attached to the SATA controller	A
SS-02	Virtual Media Manager listing storage1.vdi (20 GB) and storage2.vdi (30 GB)	A
SS-03	Output of <code>lsblk</code> showing the two new empty disks	B
SS-04	Output of <code>sudo fdisk -l</code> for the new disks only	B
SS-05	The <code>fdisk</code> session where you write the partition table on Storage 1	C
SS-06	Output of <code>sudo mkfs.ext4</code> on Storage 1's partition	C
SS-07	The <code>fdisk</code> session where you create both partitions on Storage 2	D
SS-08	The <code>fdisk p</code> command showing both partitions and their sizes	D
SS-09	Output of <code>mkfs.ext4</code> and <code>mkfs.ntfs</code> on Storage 2's two partitions	D
SS-10	The three mount point directories you created, shown with <code>ls -l /mnt</code>	E
SS-11	Output of <code>lsblk -f</code> with all three partitions mounted and their file system types visible	E
SS-12	Output of <code>df -h</code> showing all three mounted partitions	E
SS-13	The contents of your three text files, displayed with <code>cat</code>	F
SS-14	Output of <code>lsblk</code> after unmounting, with the MOUNTPOINTS column empty for all three	G

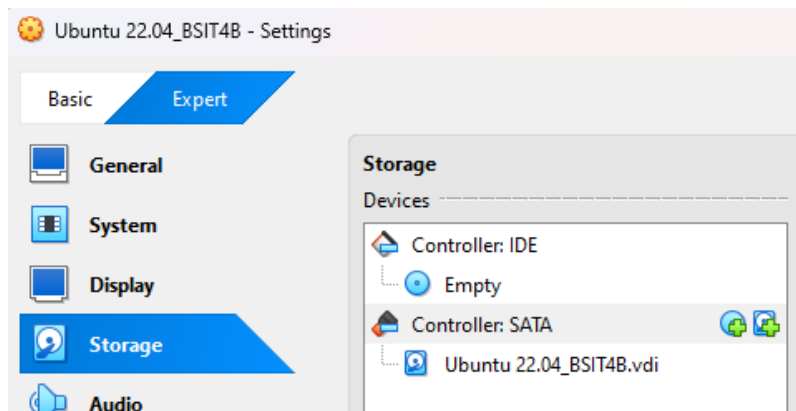
Part A — Attach the Two Virtual Disks

Time: 15 minutes | Evidence: SS-01, SS-02

A virtual disk in VirtualBox is a file on your host machine (.vdi) that the guest treats as if it were real hardware. Creating one is the equivalent of screwing a physical drive into a server chassis — the operating system will not do anything with it until you partition and format it.

Steps

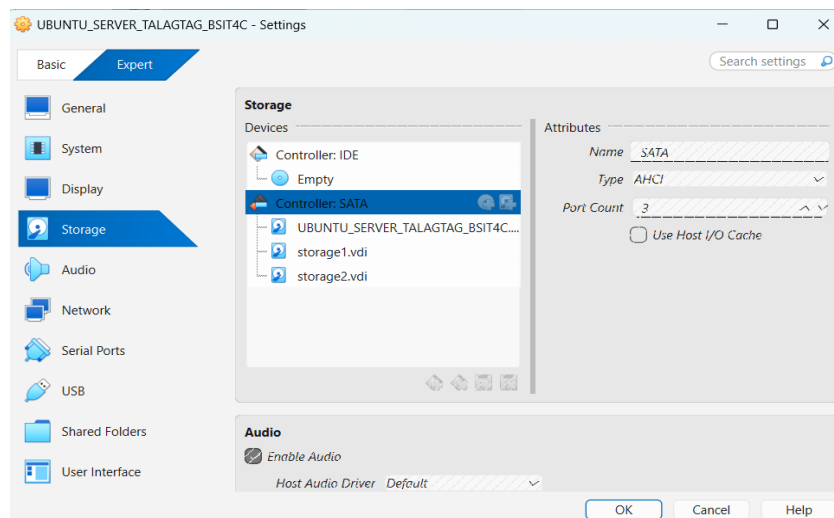
1. Make sure the VM is completely powered off. Not saved state — powered off.
2. Select the VM in VirtualBox Manager and click Settings, then Storage.
3. Click on Controller: SATA, then click the small hard-disk icon with the green plus sign (Add Hard Disk).
4. Choose Create, keep VDI as the type, keep Dynamically allocated, and set the size to 20 GB. Name the file storage1.
5. Repeat steps 3 and 4 to add a second disk of 30 GB. Name the file storage2.
6. Confirm the Storage tree now lists three disks under Controller: SATA — your original Ubuntu disk plus storage1.vdi and storage2.vdi.



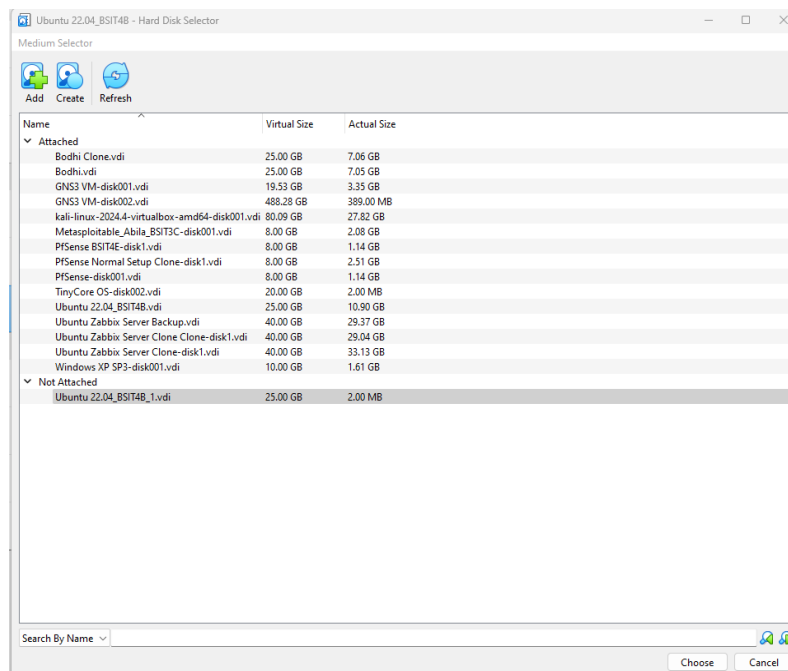
The Storage page of the VM settings. New disks are added to the SATA controller.

CAPTURE SS-01 NOW

With the Storage page open and both new disks visible in the tree, take your screenshot.



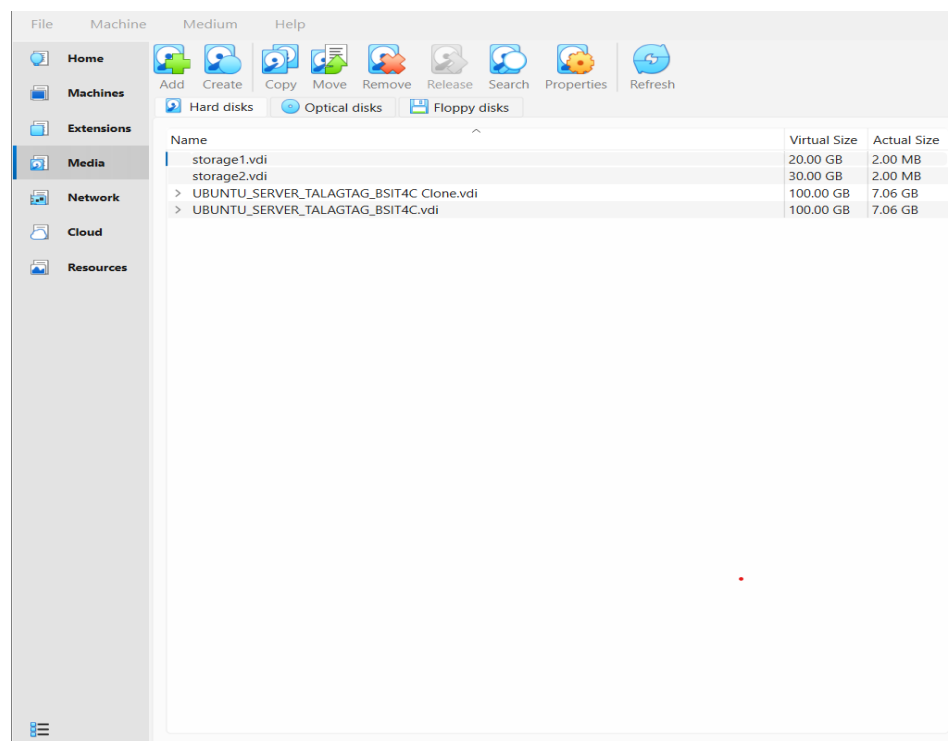
Now open File → Virtual Media Manager from the main VirtualBox window. Both new .vdi files should be listed with their virtual sizes.



The Virtual Media Manager lists every virtual disk known to VirtualBox.

CAPTURE SS-02 NOW

Screenshot the Virtual Media Manager showing storage1.vdi at 20 GB and storage2.vdi at 30 GB.



Record what you did

Item	Your answer
Full path of storage1.vdi on the host	
Full path of storage2.vdi on the host	
Allocation type chosen (dynamic / fixed)	
Controller the disks were attached to	

Part B — Boot Up and Identify the New Disks

Time: 10 minutes | Evidence: SS-03, SS-04

Start the VM and log in. The disks are attached, but Linux has not been told what to do with them yet — they exist as device files under `/dev` and nothing more.

```
ubuntu-server: ~
whoami
lsblk
sudo fdisk -l | less
```

In the `lsblk` output you are looking for two devices with the right sizes and, importantly, with nothing listed underneath them. A disk with no child entries has no partition table.

```
ian@ian-VirtualBox:/dev$ ls sd*
sda  sda1  sda2  sda3
```

Disks and partitions appear as files under `/dev` — `sda` is the first drive, `sda1` is its first partition.

READ THE NAMES CAREFULLY

On almost every setup your existing system disk is `/dev/sda`, the 20 GB disk is `/dev/sdb` and the 30 GB disk is `/dev/sdc`. **Do not assume this. Match by SIZE in the `lsblk` output, then write the confirmed names into the table below and use your own names for the rest of the task.**

Confirm your device names

Disk	Size shown by lsblk	Device name on YOUR machine	Partitions listed?
Existing system disk			
Storage 1 (20 GB)			
Storage 2 (30 GB)			

CAPTURE SS-03 AND SS-04 NOW

SS-03: the full `lsblk` output. SS-04: the `fdisk -l` section for the two new disks, showing the message that they do not contain a recognised partition table.

```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

* Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.

  https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

71 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Wed Sep 23 05:22:55 UTC 2026 on tty1
jennylyn@srv-core:~$ whoami
jennylyn
jennylyn@srv-core:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 115.3M 1 loop /snap/lxd/40575
loop1 7:1 0 87M 1 loop /snap/lxd/29351
loop2 7:2 0 63.8M 1 loop /snap/core20/2922
loop3 7:3 0 38.8M 1 loop /snap/snapd/21759
loop4 7:4 0 74M 1 loop /snap/core22/2955
loop5 7:5 0 63.9M 1 loop /snap/core20/2318
sda 8:0 0 100G 0 disk
├─sda1 8:1 0 1M 0 part
└─sda2 8:2 0 100G 0 part /
sdb 8:16 0 20G 0 disk
sdc 8:32 0 30G 0 disk
sr0 11:0 1 1024M 1 rom
jennylyn@srv-core:~$ _
```

```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop5: 63.95 MiB, 67051520 bytes, 130960 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sda: 100 GiB, 107374182400 bytes, 209715200 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 3384D5CB-6E93-48D6-87FA-5D0A5C03992B

Device Start End Sectors Size Type
/dev/sda1 2048 4095 2048 1M BIOS boot
/dev/sda2 4096 209713151 209709056 100G Linux filesystem

Disk /dev/sdb: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdc: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
jennylyn@srv-core:~$
```

Part C — Storage 1: One Partition, Formatted ext4

Time: 20 minutes | Evidence: SS-05, SS-06

Storage 1 gets the simplest possible layout: one partition using the whole 20 GB, formatted ext4. This becomes the company archive.

C.1 Create the partition table and the partition

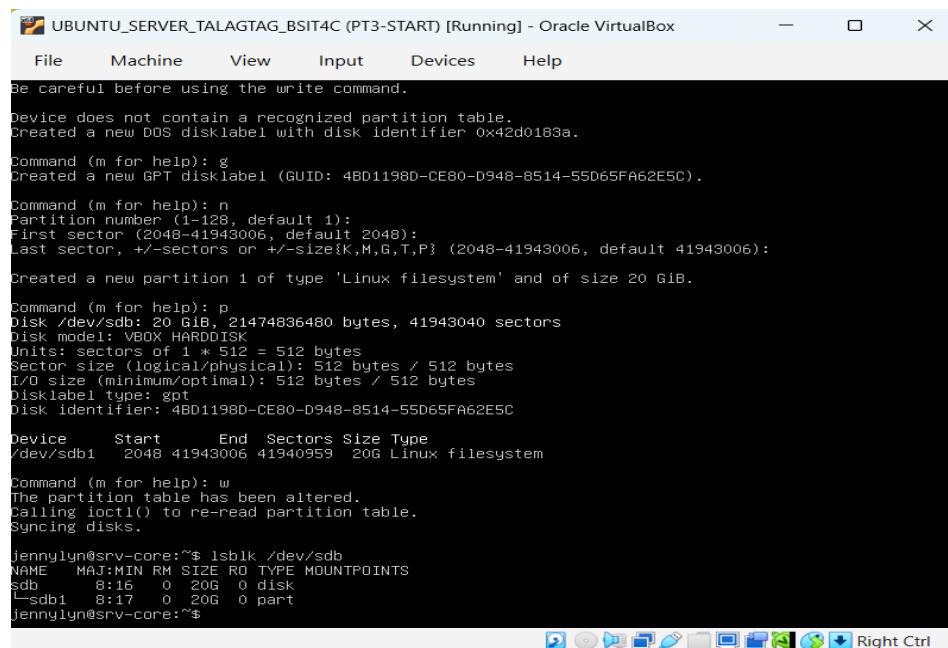
```
ubuntu-server: ~  
# use your own device name if it differs from sdb  
sudo fdisk /dev/sdb
```

Inside fdisk, type these one at a time, pressing Enter after each:

Type this	What it does
g	Create a new empty GPT partition table (this wipes the disk)
n	Add a new partition
(Enter) three times	Accept partition number 1, the default first sector, and the default last sector — which uses the entire disk
p	Print the table so you can check your work before committing
w	Write the table to disk and exit
q	Quit WITHOUT saving — use this whenever you make a mistake

CAPTURE SS-05 NOW

Screenshot the fdisk session — the commands you typed plus the confirmation that the partition table has been written. Then run `lsblk /dev/sdb` to confirm the new partition exists.



```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox  
File Machine View Input Devices Help  
Be careful before using the write command.  
Device does not contain a recognized partition table.  
Created a new DOS disklabel with disk identifier 0x42d0183a.  
Command (m for help): g  
Created a new GPT disklabel (GUID: 4BD1198D-CE80-D948-8514-55D65FA62E5C).  
Command (m for help): n  
Partition number (1-128, default 1):  
First sector (2048-41943006, default 2048):  
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943006, default 41943006):  
Created a new partition 1 of type 'Linux filesystem' and of size 20 GiB.  
Command (m for help): p  
Disk /dev/sdb: 20 GiB, 21474836480 bytes, 41943040 sectors  
Disk model: VBOX HARDDISK  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
Disklabel type: gpt  
Disk identifier: 4BD1198D-CE80-D948-8514-55D65FA62E5C  


| Device    | Start | End      | Sectors  | Size | Type             |
|-----------|-------|----------|----------|------|------------------|
| /dev/sdb1 | 2048  | 41943006 | 41940959 | 20G  | Linux filesystem |

  
Command (m for help): w  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
Jennylyn@srv-core:~$ lsblk /dev/sdb  
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS  
sdb 8:16 0 20G 0 disk  
└─sdb1 8:17 0 20G 0 part  
Jennylyn@srv-core:~$
```

C.2 Format the partition as ext4

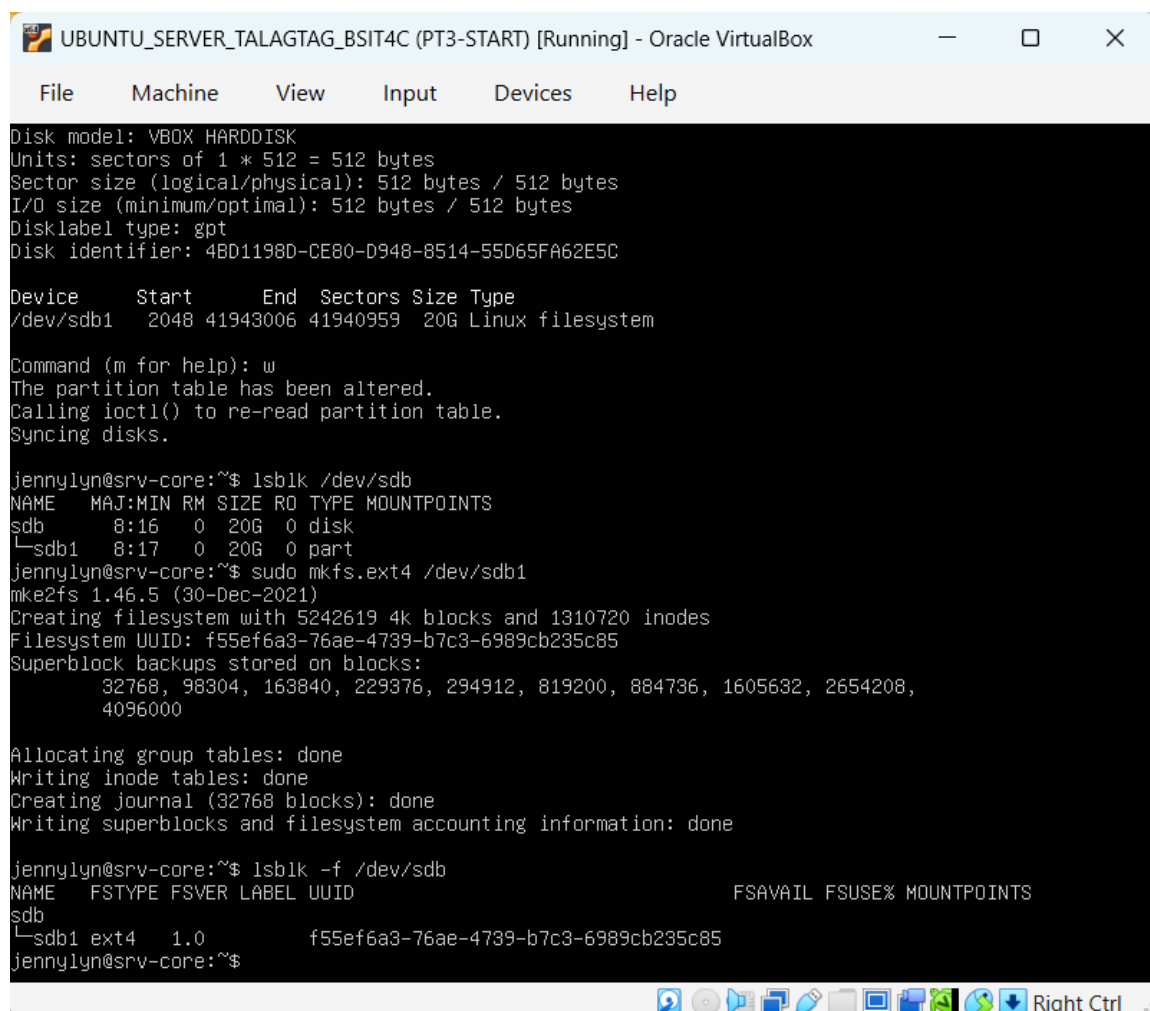
A partition is only a reserved region of the disk. Formatting installs a file system into that region. Until you format it, the partition cannot hold a single file.

```
ubuntu-server: ~  
sudo mkfs.ext4 /dev/sdb1  
  
# then confirm the file system type is now visible:  
lsblk -f /dev/sdb
```

Partition device name	Size reported by lsblk	First 8 characters of the UUID
/dev/sdb1	20GB	f55ef6a3

CAPTURE SS-06 NOW

The mkfs.ext4 output plus the lsblk -f result showing ext4 in the FSTYPE column.



```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox  
File Machine View Input Devices Help  
Disk model: VBOX HARDDISK  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
Disklabel type: gpt  
Disk identifier: 4BD1198D-CE80-D948-8514-55D65FA62E5C  
  
Device      Start      End  Sectors Size Type  
/dev/sdb1   2048 41943006 41940959  20G Linux filesystem  
  
Command (m for help): w  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
  
jennylyn@srv-core:~$ lsblk /dev/sdb  
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS  
sdb    8:16  0  20G  0 disk  
└─sdb1  8:17  0  20G  0 part  
jennylyn@srv-core:~$ sudo mkfs.ext4 /dev/sdb1  
mke2fs 1.46.5 (30-Dec-2021)  
Creating filesystem with 5242619 4k blocks and 1310720 inodes  
Filesystem UUID: f55ef6a3-76ae-4739-b7c3-6989cb235c85  
Superblock backups stored on blocks:  
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,  
        4096000  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (32768 blocks): done  
Writing superblocks and filesystem accounting information: done  
  
jennylyn@srv-core:~$ lsblk -f /dev/sdb  
NAME FSTYPE FSVER LABEL UUID                               FSAVAIL FSUSE% MOUNTPOINTS  
sdb  
└─sdb1 ext4    1.0             f55ef6a3-76ae-4739-b7c3-6989cb235c85  
jennylyn@srv-core:~$
```

Part D — Storage 2: Two Partitions, Two File Systems

Time: 35 minutes | Evidence: SS-07, SS-08, SS-09

This is the part that separates a technician from someone who can only follow a wizard. One physical disk, split into two logical regions, each carrying a different file system, each serving a different purpose.

D.1 Install NTFS support

Ubuntu Server does not ship with the tool that creates NTFS file systems. Install it now, before you need it.

```
ubuntu-server: ~  
sudo apt update  
sudo apt install -y ntfs-3g  
  
# verify the formatting tool is present:  
which mkfs.ntfs
```

D.2 Create both partitions

Start fdisk against **/dev/sdc** — again, use your own confirmed device name.

```
ubuntu-server: ~  
sudo fdisk /dev/sdc
```

Type this	What it does
g	New empty GPT partition table
n	Add the first partition
(Enter)	Accept partition number 1
(Enter)	Accept the default first sector
+15G	Make this partition exactly 15 GB
n	Add the second partition
(Enter)	Accept partition number 2
(Enter)	Accept the default first sector
(Enter)	Accept the default last sector — takes all remaining space
t	Change a partition type
2	Apply the change to partition 2
L	List the available type codes and find Microsoft basic data
(its number)	Enter the number shown next to Microsoft basic data — usually 11
p	Print the table and check both sizes before committing
w	Write and exit

WHY CHANGE THE PARTITION TYPE?

The type code is a label stored in the partition table. It does not format anything and it does not stop Linux from using the partition. What it does is tell other operating systems what to expect. Set it correctly and a Windows machine recognises the partition immediately; leave it as Linux filesystem and Windows may offer to format it instead.

YOUR SECOND PARTITION WILL NOT BE EXACTLY 15 GB

Expect something close to 14.9 GB. That is correct behaviour, not an error, and Concept Question 3 asks you to explain it. Do not go back and try to force the numbers to match.

D.3 Record the layout before you format

Partition	Device name	Size from fdisk p	Type code shown
Storage 2, partition 1	/dev/sdc1	15 GiB	Linux filesystem
Storage 2, partition 2	/dev/sdc2	15 GiB	Microsoft basic data

CAPTURE SS-07 AND SS-08 NOW

SS-07: the fdisk session with both partitions being created. SS-08: the output of the p command showing both partitions, their sizes and their type codes.

```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Partition number (2-128, default 2):
First sector (31459328-62914526, default 31459328):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-62914526, default 62914526): ~15 GB
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-62914526, default 62914526): ~15GB
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-62914526, default 62914526): ~15gGB
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-62914526, default 62914526):
Created a new partition 2 of type 'Linux filesystem' and of size 15 GiB.
Command (m for help):
Command (m for help): t
Partition number (1,2, default 2): 2
Partition type or alias (type L to list all): 11
Changed type of partition 'Linux filesystem' to 'Microsoft basic data'.
Command (m for help): p
Disk /dev/sdc: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 9030D9FE-CB45-A048-9682-E6CBD22D5356

Device      Start      End  Sectors Size Type
/dev/sdc1   2048 31459327 31457280 15G Linux filesystem
/dev/sdc2 31459328 62914526 31455199 15G Microsoft basic data

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

jennylyn@srv-core:~$ _
```

```
jennylyn@srv-core:~$ sudo mkfs.ntfs -f /dev/sdc1
Cluster size has been automatically set to 4096 bytes.
Creating NTFS volume structures.
mkntfs completed successfully. Have a nice day.
jennylyn@srv-core:~$ sudo mkfs.ntfs -f /dev/sdc2
Cluster size has been automatically set to 4096 bytes.
Creating NTFS volume structures.
mkntfs completed successfully. Have a nice day.
jennylyn@srv-core:~$ lsblk -f
```

NAME	FSTYPE	FSVER	LABEL	UUID	FSAVAIL	FSUSE%	MOUNTPOINTS
loop0	squashfs	4.0			0	100%	/snap/lxd/40575
loop1	squashfs	4.0			0	100%	/snap/lxd/29351
loop2	squashfs	4.0			0	100%	/snap/core20/2922
loop3	squashfs	4.0			0	100%	/snap/snapd/21759
loop4	squashfs	4.0			0	100%	/snap/core22/2955
loop5	squashfs	4.0			0	100%	/snap/core20/2318
sda							
└─sda1							
└─sda2	ext4	1.0		46e3089b-b650-46ff-9a3a-2e33dd0ec43d	85.3G	8%	/
sdb							
└─sdb1	ext4	1.0		f55ef6a3-76ae-4739-b7c3-6989cb235c85			
sdc							
└─sdc1	ntfs			6234CA4D2E6A5D53			
└─sdc2	ntfs			47C823347724F3E5			
sr0							

```
jennylyn@srv-core:~$ _
```

D.4 Format each partition with its own file system

```
ubuntu-server: ~
# Partition 1 -> ext4, for Finance records
sudo mkfs.ext4 /dev/sdc1

# Partition 2 -> NTFS, for the Windows exchange folder
sudo mkfs.ntfs -f -L IVT_EXCHANGE /dev/sdc2

# confirm both:
lsblk -f
```

THE -f FLAG IS IMPORTANT

Without **-f**, mkfs.ntfs performs a full format and zeroes every sector on the partition. On 15 GB that can take twenty minutes or more, and you do not have twenty minutes. The **-f** flag performs a quick format instead, which is what you want here. **-L IVT_EXCHANGE** sets a volume label so the partition is easy to recognise later.

CAPTURE SS-09 NOW

One screenshot containing both format commands and the lsblk -f output that shows ext4 on partition 1 and ntfs on partition 2.

```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: 6ca1b526-1f4f-4322-b8d7-056a3bee6141
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

jennylyn@srv-core:~$ sudo mkfs.ntfs -f -L IVT_EXHANGE /dev/sdc2
Cluster size has been automatically set to 4096 bytes.
Creating NTFS volume structures.
mkntfs completed successfully. Have a nice day.
jennylyn@srv-core:~$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                FSAVAIL FSUSE% MOUNTPOINTS
loop0 squash 4.0                                     0      100% /snap/1xd/40575
loop1 squash 4.0                                     0      100% /snap/1xd/29351
loop2 squash 4.0                                     0      100% /snap/core20/2922
loop3 squash 4.0                                     0      100% /snap/snapd/21759
loop4 squash 4.0                                     0      100% /snap/core22/2955
loop5 squash 4.0                                     0      100% /snap/core20/2318
sda
├─ sda1
├─ sda2
├─ sdb
│ └─ sdb1
│     └─ ext4      1.0                                f55ef6a3-76ae-4739-b7c3-6989cb235c85
└─ sdc
    ├─ sdc1
    │ └─ ext4      1.0                                6ca1b526-1f4f-4322-b8d7-056a3bee6141
    └─ sdc2
        └─ ntfs     IVT_EXHANGE 4C1D29712C1B0588
sr0
jennylyn@srv-core:~$ _
```

Part E — Create Mount Points and Mount Everything

Time: 20 minutes | Evidence: SS-10, SS-11, SS-12

A formatted partition is still unreachable. Linux has no drive letters — there is no D: to click on. Instead you attach the partition to an empty directory somewhere in the single tree that starts at /. That directory is the mount point, and after mounting, everything you write inside it lands on the partition rather than on the system disk.

E.1 Decide and create your mount points

Use **/mnt** — the directory that exists specifically for storage an administrator mounts by hand. (**/media** is for removable devices the system mounts automatically.) Name each directory after the department it serves.

```
ubuntu-server: ~
sudo mkdir -p /mnt/ivt_archive
sudo mkdir -p /mnt/ivt_finance
sudo mkdir -p /mnt/ivt_exchange

ls -l /mnt
```

CAPTURE SS-10 NOW

Screenshot the ls -l /mnt output showing all three empty directories.

```

UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Writing superblocks and filesystem accounting information: done

jennylyn@srv-core:~$ sudo mkfs.ntfs -f -L IVT_EXCHANGE /dev/sdc2
Cluster size has been automatically set to 4096 bytes.
Creating NTFS volume structures.
mkntfs completed successfully. Have a nice day.
jennylyn@srv-core:~$ lsblk -f
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE% MOUNTPPOINTS
loop0 squash 4.0 0 100% /snap/lxd/40575
loop1 squash 4.0 0 100% /snap/lxd/29351
loop2 squash 4.0 0 100% /snap/core20/2922
loop3 squash 4.0 0 100% /snap/snapd/21759
loop4 squash 4.0 0 100% /snap/core22/2955
loop5 squash 4.0 0 100% /snap/core20/2318
sda
├─sda1
├─sda2
└─sdb
   └─sdb1
      ext4 1.0 46e3089b-b650-46ff-9a3a-2e33dd0ec43d 85.3G 8% /
sdc
├─sdc1
│   ext4 1.0 f55ef6a3-76ae-4739-b7c3-6989cb235c85
└─sdc2
    ntfs IVT_EXCHANGE 4C1D29712C1B0588
sr0
jennylyn@srv-core:~$ sudo mkdir -p /mnt/ivt_archive
jennylyn@srv-core:~$ sudo mkdir -p /mnt/ivt_finance
jennylyn@srv-core:~$ sudo mkdir -p /mnt/ivt_exchange
jennylyn@srv-core:~$ ls -l /mnt
total 12
drwxr-xr-x 2 root root 4096 Sep 23 07:02 ivt_archive
drwxr-xr-x 2 root root 4096 Sep 23 07:03 ivt_exchange
drwxr-xr-x 2 root root 4096 Sep 23 07:02 ivt_finance
jennylyn@srv-core:~$

```

E.2 Mount all three partitions

```

ubuntu-server: ~
sudo mount /dev/sdb1 /mnt/ivt_archive
sudo mount /dev/sdc1 /mnt/ivt_finance
sudo mount /dev/sdc2 /mnt/ivt_exchange

lsblk -f
df -h

```

The mount command produces no output when it succeeds. Silence means it worked. If you get an error, read it before doing anything else — most mount failures name the exact problem.

E.3 Fill in your mount map

Partition	Device	Mount point	File system	Size in df -h	✓
Company archive	/dev/sdb1	/mnt/ivt_archive	ext4	20G	/
Finance records	/dev/sdc1	/mnt/ivt_finance	ext4	15G	/

Partition	Device	Mount point	File system	Size in df -h	✓
Windows exchange	/dev/sdc2	/mnt/ivt_exchange	ntfs	15G	/

CAPTURE SS-11 AND SS-12 NOW

SS-11: lsblk -f with all three file system types and mount points. SS-12: df -h with all three mounted.

```
jennylyn@srv-core:~$ sudo umount /mnt/ivt_exchange
[sudo] password for jennylyn:
jennylyn@srv-core:~$ sudo umount /mnt/ivt_finance
jennylyn@srv-core:~$ sudo umount /mnt/ivt_archive
jennylyn@srv-core:~$ sudo mount /dev/sdb1 /mnt/ivt_archive
jennylyn@srv-core:~$ sudo mount /dev/sdc1 /mnt/ivt_finance
jennylyn@srv-core:~$ sudo mount /dev/sdc2 /mnt/ivt_exchange
jennylyn@srv-core:~$ lsblk -f
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE% MOUNTPOINTS
loop0 squash 4.0 0 100% /snap/lxd/40575
loop1 squash 4.0 0 100% /snap/lxd/29351
loop2 squash 4.0 0 100% /snap/core20/2922
loop3 squash 4.0 0 100% /snap/snapd/21759
loop4 squash 4.0 0 100% /snap/core22/2955
loop5 squash 4.0 0 100% /snap/core20/2318
sda
├─sda1
└─sda2
   ext4 1.0 46e3089b-b650-46ff-9a3a-2e33dd0ec43d 85.3G 8% /
sdb
└─sdb1
   ext4 1.0 f55ef6a3-76ae-4739-b7c3-6989cb235c85 18.5G 0% /mnt/ivt_archive
sdc
├─sdc1
│  ext4 1.0 6ca1b526-1f4f-4322-b8d7-056a3bee6141 13.9G 0% /mnt/ivt_finance
└─sdc2
   ntfs IVT_EXHANGE 4C1D29712C1B0588 14.9G 0% /mnt/ivt_exchange
sr0
jennylyn@srv-core:~$ _
```


UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
— □ ×

File Machine View Input Devices Help

```
jennylyn@srv-core:~$ sudo umount /mnt/ivt_archive
jennylyn@srv-core:~$ sudo mount /dev/sdb1 /mnt/ivt_archive
jennylyn@srv-core:~$ sudo mount /dev/sdc1 /mnt/ivt_finance
jennylyn@srv-core:~$ sudo mount /dev/sdc2 /mnt/ivt_exchange
jennylyn@srv-core:~$ lsblk -f
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE% MOUNTPOINTS
loop0 squash 4.0 0 100% /snap/lxd/40575
loop1 squash 4.0 0 100% /snap/lxd/29351
loop2 squash 4.0 0 100% /snap/core20/2922
loop3 squash 4.0 0 100% /snap/snapd/21759
loop4 squash 4.0 0 100% /snap/core22/2955
loop5 squash 4.0 0 100% /snap/core20/2318
sda
├─sda1
└─sda2
   ext4 1.0 46e3089b-b650-46ff-9a3a-2e33dd0ec43d 85.3G 8% /
sdb
└─sdb1
   ext4 1.0 f55ef6a3-76ae-4739-b7c3-6989cb235c85 18.5G 0% /mnt/ivt_archive
sdc
├─sdc1
│  ext4 1.0 6ca1b526-1f4f-4322-b8d7-056a3bee6141 13.9G 0% /mnt/ivt_finance
└─sdc2
   ntfs IVT_EXHANGE 4C1D29712C1B0588 14.9G 0% /mnt/ivt_exchange
sr0
jennylyn@srv-core:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            392M  1.1M  391M   1% /run
/dev/sda2        98G   7.6G   86G   9% /
tmpfs            2.0G   0 2.0G   0% /dev/shm
tmpfs            5.0M   0 5.0M   0% /run/lock
tmpfs            392M  4.0K  392M   1% /run/user/1000
/dev/sdb1         20G   32K   19G   1% /mnt/ivt_archive
/dev/sdc1         15G   28K   14G   1% /mnt/ivt_finance
/dev/sdc2         15G   65M   15G   1% /mnt/ivt_exchange
jennylyn@srv-core:~$
```


Right Ctrl

Part F — Write One File to Each Partition

Time: 15 minutes | Evidence: SS-13

Mounting proves the partition is reachable. Writing a file proves it is usable. Create one text file on each of the three partitions, each containing your own identifying details so the file is unmistakably yours.

F.1 Create the three files

Use **cat** > to type content directly into a new file. Press **Ctrl+D** on a line of its own when you are finished — that signals end of input.

```
ubuntu-server: ~
sudo cat > /mnt/ivt_archive/archive_note.txt
Isla Verde Trading Corp - Company Archive
Prepared by: <your full name>
Section: <your section>      Date: <today>
Device: /dev/sdb1   File system: ext4
[press Ctrl+D here]
```

Repeat for the other two partitions, changing the device, file system and folder each time:

```
ubuntu-server: ~
sudo cat > /mnt/ivt_finance/finance_note.txt
sudo cat > /mnt/ivt_exchange/exchange_note.txt
```

F.2 Verify all three files

```
ubuntu-server: ~
ls -l /mnt/ivt_archive /mnt/ivt_finance /mnt/ivt_exchange

cat /mnt/ivt_archive/archive_note.txt
cat /mnt/ivt_finance/finance_note.txt
cat /mnt/ivt_exchange/exchange_note.txt
```

IF PERMISSION IS DENIED

A freshly formatted ext4 partition is owned by root, so an ordinary user cannot write into it. You can either keep using **sudo** as shown above, or hand the directory to yourself with **sudo chown \$USER:\$USER /mnt/ivt_archive**. Ownership and permissions are the subject of Weeks 7 and 8 — for today, sudo is enough.

CAPTURE SS-13 NOW

One screenshot showing the contents of all three files displayed with cat, with your name visible in each.

```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

sdsc1 8:33 0 15G 0 part /mnt/ivt_finance
sdsc2 8:34 0 15G 0 part /mnt/ivt_exchange
sr0 11:0 1 1024M 1 rom
jennylyn@srv-core:~$ echo "IVT Archive - 20GB Storage 1" | sudo tee /mnt/ivt_archive/archive_note.txt
IVT Archive - 20GB Storage 1
jennylyn@srv-core:~$ echo "Finance - 15GB ext4 Partion" | sudo tee /mnt/ivt_finance/finance_note.txt
Finance - 15GB ext4 Partion
jennylyn@srv-core:~$ echo "Exchange - 15GB Partion" | sudo tee /mnt/ivt_exchange/exchange_note.txt
Exchange - 15GB Partion
jennylyn@srv-core:~$ ls -l /mnt/ivt_archive/[12932.191854] watchdog: BUG: soft lockup - CPU#0 stuck
for 2048s! [swapper/0:0]
[17197.890368] watchdog: BUG: soft lockup - CPU#0 stuck for 2310s! [swapper/0:0]
[20839.257209] watchdog: BUG: soft lockup - CPU#0 stuck for 3388s! [appont:1863]
[20840.624731] systemd[1]: Failed to start Network Name Resolution.
[20840.631852] systemd[1]: Failed to start Rule-based Manager for Device Events and Files.

total 24
-rw-r--r-- 1 root root 29 Sep 23 08:18 archive_note.txt
-rw-r--r-- 1 root root 35 Sep 23 07:58 exchange_note.txt
drwx----- 2 root root 16384 Sep 23 05:45 lost+found
jennylyn@srv-core:~$ ls -l /mnt/ivt_archive/ /mnt/ivt_finance/ /mnt/ivt_exchange
/mnt/ivt_archive/:
total 24
-rw-r--r-- 1 root root 29 Sep 23 08:18 archive_note.txt
-rw-r--r-- 1 root root 35 Sep 23 07:58 exchange_note.txt
drwx----- 2 root root 16384 Sep 23 05:45 lost+found

/mnt/ivt_exchange:
total 1
-rwxrwxrwx 1 root root 24 Sep 23 08:22 exchange_note.txt

/mnt/ivt_finance/:
total 20
-rw-r--r-- 1 root root 28 Sep 23 08:19 finance_note.txt
drwx----- 2 root root 16384 Sep 23 06:53 lost+found
jennylyn@srv-core:~$
```

Part G — Unmount Everything Cleanly

Time: 10 minutes | Evidence: SS-14

When you write a file, Linux does not necessarily push it to the disk immediately. It holds data in memory and writes it out when convenient, because that is dramatically faster. Unmounting forces every pending write to complete and then detaches the file system. This is why Linda's thesis went missing — the USB was pulled before the write finished.

```
ubuntu-server: ~
# make sure your shell is NOT sitting inside any mount point:
cd /

sudo umount /mnt/ivt_archive
sudo umount /mnt/ivt_finance
sudo umount /mnt/ivt_exchange

lsblk
df -h
```

"TARGET IS BUSY"

Something is still using the file system — usually your own shell sitting in that directory. Run `cd /` and try again. To find the culprit, use `lsuf +D /mnt/ivt_archive`. Never force an unmount just to make the message go away — that is how data gets corrupted.

Final verification

Check	Expected result	✓
lsblk MOUNTPOINTS for sdb1, sdc1, sdc2	All three empty	☐
df -h	None of the three partitions appear	☐
ls /mnt/ivt_archive	Empty — the directory still exists but the file is not visible	☐
FSTYPE column in lsblk -f	Still shows ext4 / ext4 / ntfs	☐

THE LAST CHECK IS THE MOST IMPORTANT ONE

After unmounting, the mount point directory is empty — but your file has not been deleted. Understanding why is the whole point of this task, and Concept Question 7 asks you to explain it.

CAPTURE SS-14 NOW

Screenshot lsblk after unmounting, with the MOUNTPOINTS column empty for all three partitions.

```
UBUNTU_SERVER_TALAGTAG_BSIT4C (PT3-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
/dev/sdb1 20G 32K 19G 1% /mnt/ivt_archive
/dev/sdc1 15G 28K 14G 1% /mnt/ivt_finance
/dev/sdc2 15G 65M 15G 1% /mnt/ivt_exchange
jennylyn@srv-core:~$ cd /
jennylyn@srv-core:/$ sudo umount /mnt/ivt_archive
jennylyn@srv-core:/$ sudo umount /mnt/ivt_finance
jennylyn@srv-core:/$ sudo umount /mnt/ivt_exchange
jennylyn@srv-core:/$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 115.3M 1 loop /snap/lxd/40575
loop1 7:1 0 87M 1 loop /snap/lxd/29351
loop2 7:2 0 63.8M 1 loop /snap/core20/2922
loop3 7:3 0 38.8M 1 loop /snap/snapd/21759
loop4 7:4 0 74M 1 loop /snap/core22/2955
loop5 7:5 0 63.9M 1 loop /snap/core20/2318
sda 8:0 0 100G 0 disk
├─sda1 8:1 0 1M 0 part
└─sda2 8:2 0 100G 0 part /
sdb 8:16 0 20G 0 disk
└─sdb1 8:17 0 20G 0 part
sdc 8:32 0 30G 0 disk
├─sdc1 8:33 0 15G 0 part
└─sdc2 8:34 0 15G 0 part
sr0 11:0 1 1024M 1 rom
jennylyn@srv-core:/$ df -h
Filesystem Size Used Avail Use% Mounted on
tmpfs 392M 1.1M 391M 1% /run
/dev/sda2 98G 7.6G 86G 9% /
tmpfs 2.0G 0 2.0G 0% /dev/shm
tmpfs 5.0M 0 5.0M 0% /run/lock
tmpfs 392M 4.0K 392M 1% /run/user/1000
```

```
jennylyn@srv-core:/$ ls /mnt/ivt_archive
jennylyn@srv-core:/$ lsblk -f
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE% MOUNTPOINTS
loop0 squash 4.0 0 100% /snap/lxd/40575
loop1 squash 4.0 0 100% /snap/lxd/29351
loop2 squash 4.0 0 100% /snap/core20/2922
loop3 squash 4.0 0 100% /snap/snapd/21759
loop4 squash 4.0 0 100% /snap/core22/2955
loop5 squash 4.0 0 100% /snap/core20/2318
sda
├─sda1
├─sda2
│   └─ ext4 1.0 46e3089b-b650-46ff-9a3a-2e33dd0ec43d 85.3G 8% /
├─sdb
│   └─sdb1
│       └─ ext4 1.0 f55ef6a3-76ae-4739-b7c3-6989cb235c85
├─sdc
│   ├──sdc1
│   │   └─ ext4 1.0 6ca1b526-1f4f-4322-b8d7-056a3bee6141
│   └─sdc2
│       └─ ntfs IVT_EXCHANGE 4C1D29712C1B0588
sr0
jennylyn@srv-core:/$
```

```
ubuntu-server: ~
```

```
# finally, shut the VM down properly, then take a snapshot named PT3-DONE
```

```
sudo shutdown -h now
```

Part H — Concept Questions

Time: 15 minutes | 30 of the 100 points

Answer in your own words, in complete sentences, in the space provided under each question. Three to five sentences is the right length — enough to show reasoning, short enough to stay focused. Answers copied from the slides or from a classmate earn zero for the entire section.

- Q1. Linux gave your new disks the names `sdb` and `sdc`, and the partitions on the second one became `sdc1` and `sdc2`. Break down the name `/dev/sdc2` piece by piece and explain what each part tells you. Then explain why Linux uses this scheme instead of the drive letters `D:` and `E:` that Windows would have assigned.

-
- Q2. You created a partition, and then you formatted it. Explain the difference between these two operations. Include in your answer what would happen if you tried to mount a partition that had been created but never formatted.

-
- Q3. Storage 2 is a 30 GB disk and you asked for a 15 GB first partition, yet the second partition came out slightly under 15 GB. Explain where the missing space went. Your answer should mention at least two things that consume space on a disk before any user file is written.

-
- Q4. Your instructor asks why the exchange partition uses NTFS when everything else on the server is `ext4`. Give the argument in favour of the NTFS choice for that specific partition, then give one real disadvantage the Isla Verde team accepts by making it. Be specific about what `ext4` provides that NTFS-on-Linux does not.

-
- Q5. A colleague suggests you skip creating `/mnt/ivt_finance` and instead mount the Finance partition directly onto `/home`, since that directory already exists and already has files in it. Explain what would happen to the existing contents of `/home`, and why mount points should normally be empty directories.
-

Q6. Linda saved her thesis in Ubuntu, could not find it in Windows, and later got a "No such file or directory" error when the filename looked correct to her. Identify the two separate technical causes behind these two symptoms, and state which of them your NTFS partition is designed to prevent.

Q7. After you unmounted everything in Part G, the command `ls /mnt/ivt_archive` returned nothing — yet the text file you created has not been deleted. Explain where the file actually is and what mounting does. Then state what you would need to type to see the file again.

Q8. Vince, the intern from the case study, was told the server was out of space. He guessed which directory was the problem and deleted files from `/bin`, which broke the server completely. Name the single command he should have run first, state exactly what information it would have given him, and explain how that information would have changed his decision.

Q9. You try to unmount a partition and get the message "target is busy". Explain what that message means, describe the correct procedure for resolving it, and explain the specific risk of using a forced unmount to bypass it instead.

Q10. Reboot the VM and check `/mnt` with `lsblk`. Your three partitions are no longer mounted, even though you never touched them. Explain why mounts do not survive a reboot, and name the configuration file an administrator would edit to make them permanent.

5 What to Submit

One Microsoft Word document (.docx). Nothing else — no zip archives, no loose image files, no PDFs.

File name

Naming convention

PT3_FileSystem_<LASTNAME>_<FIRSTNAME>_<SECTION>.docx

example:

PT3_FileSystem_DELACRUZ_JUAN_BSIT4A.docx

Required contents, in this order

#	Section	Must contain
1	Cover page	Your name, section, date performed, date submitted, title of the task
2	Part A	SS-01, SS-02, and the completed disk record table
3	Part B	SS-03, SS-04, and the completed device name table
4	Part C	SS-05, SS-06, and the completed Storage 1 table
5	Part D	SS-07, SS-08, SS-09, and the completed layout table
6	Part E	SS-10, SS-11, SS-12, and the completed mount map
7	Part F	SS-13
8	Part G	SS-14 and the completed final verification checklist
9	Part H	All ten concept questions answered
10	Declaration	The signed statement below

Declaration

I performed this task myself on my own virtual machine. Every screenshot in this document was captured from my own screen during my own session. I understand that submitting another student's screenshots, or output that has been retyped rather than captured, is academic dishonesty and results in a grade of zero for the entire task.

Signature over printed name

Date

Deadline and late work

When submitted	Maximum score
On or before the deadline	100%
Up to 24 hours late	90%
24 to 72 hours late	75%
More than 72 hours late	Not accepted without an approved excuse

Appendix A — When Things Go Wrong

What you see	What it means	What to do
The new disks do not appear in lsblk after booting	The disks were attached while the VM was running, or attached to the wrong controller	Power the VM fully off, re-check Settings → Storage, boot again
fdisk: cannot open /dev/sdb: Permission denied	You forgot sudo	Re-run the command with sudo in front
Device or resource busy when running fdisk	A partition on that disk is currently mounted	Unmount it first, then run fdisk again
mkfs.ntfs: command not found	The ntfs-3g package is not installed	sudo apt update && sudo apt install -y ntfs-3g
mkfs.ntfs is taking forever	You omitted the -f flag, so it is zeroing every sector	Press Ctrl+C and re-run with -f
mount: /mnt/x: wrong fs type, bad option, bad superblock	The partition was never formatted, or you named the wrong device	Check lsblk -f — if FSTYPE is blank, format it first
mount: /mnt/x: mount point does not exist	You skipped the mkdir step or mistyped the path	Create the directory with sudo mkdir -p, then mount again
umount: target is busy	A process — probably your own shell — is inside the mount point	cd / then try again; use lsof +D <path> to identify the culprit
Permission denied when creating the text file	The freshly formatted partition is owned by root	Use sudo, or chown the directory to your user
Your partition sizes look wrong after a reboot	You are reading a different device — names can shift	Match by size and UUID with lsblk -f, not by name alone

Appendix B — Command Reference

Looking at storage

Command	What it does
lsblk	Tree of all block devices, their sizes and where they are mounted
lsblk -f	The same tree plus file system type, label and UUID
sudo fdisk -l	Detailed partition table for every disk
df -h	Space used and available on every mounted file system
mount	Every file system currently mounted, with its options
ls /dev/sd*	The raw device files for disks and partitions

Changing storage

Command	What it does
sudo fdisk /dev/sdX	Open the interactive partition editor for one disk
sudo mkfs.ext4 /dev/sdXn	Format a partition with the ext4 file system
sudo mkfs.ntfs -f /dev/sdXn	Quick-format a partition with NTFS
sudo mkdir -p /mnt/name	Create a mount point directory
sudo mount /dev/sdXn /mnt/name	Attach a partition to a mount point
sudo umount /mnt/name	Detach it and flush all pending writes

Inside fdisk

Key	Action	Key	Action
g	New GPT partition table	t	Change a partition's type code
o	New MS-DOS partition table	L	List available type codes
n	Create a new partition	p	Print the current table
d	Delete a partition	w	Write changes and exit
m	Show the help menu	q	Quit WITHOUT saving

SIZE SHORTHAND IN FDISK

When fdisk asks for the last sector you can type a size instead: **+15G** for 15 gibibytes, **+500M** for 500 mebibytes, or just press Enter to use everything that is left.

PERFORMANCE TASK 3 — SCORING RUBRIC

IT 4112 System Administration and Maintenance • File System Administration • 100 points

Criterion	Exemplary (100%)	Proficient (80%)	Developing (60%)	Beginning (40%)
Disk Provisioning & Identification 15 pts	Both disks created at exactly the correct sizes and attached to the SATA controller. Device names confirmed by size and recorded. SS-01 to SS-04 complete and readable.	Both disks attached at correct sizes. Device names identified with one minor recording error. All four screenshots present.	Disks attached but a size is wrong, or device names were assumed rather than verified. One screenshot missing.	Disks not attached correctly, or two or more screenshots missing. No evidence the OS detected the new hardware.
Partitioning Accuracy 25 pts	Storage 1 has one partition using the full disk. Storage 2 has two, the first exactly 15 GB, the second using all remaining space with the type code set to Microsoft basic data.	All partitions created with correct counts and near-correct sizes. Type code set. Minor gaps in the layout table.	Partitions created but one size is clearly wrong, or the second disk has the wrong partition count, or the type code was not changed.	Partitioning incomplete or done on the wrong device. Partition table not written, or evidence does not show the required layout.
Formatting & File Systems 20 pts	Storage 1 and Storage 2 partition 1 are ext4; Storage 2 partition 2 is NTFS with a volume label. lsblk -f confirms all three. The -f flag used correctly.	All three formatted with the correct file systems and confirmed by lsblk -f. Volume label missing or -f flag omitted.	Two of the three formatted correctly, or a file system type does not match the specification.	Formatting not performed, performed on the wrong partitions, or no evidence that any file system was created.
Mounting, Files & Clean Unmount 20 pts	Three purposefully named mount points under /mnt. All three mounted and verified in lsblk -f and df -h. An identifying text file on each, shown with cat. All unmounted cleanly and proven.	All three mounted, files written, and all unmounted. One verification step missing or one mount point poorly named.	Some partitions mounted or some files created, but the set is incomplete; or unmounting was not verified.	Mounting not achieved, files not written, or partitions left mounted with no clean-unmount evidence.
Concept Questions 15 pts	All ten answered in the student's own words with correct reasoning. Answers connect the commands performed to the concepts and to the case studies.	All ten answered. Technically sound overall with one or two shallow or partially incorrect explanations.	Seven to nine answered, or several answers restate the handout without showing understanding.	Fewer than seven answered, or answers show fundamental misunderstanding, or content is copied.
Documentation Quality 5 pts	Follows the required order exactly, all fourteen screenshots labelled, every record table completed, correct file name, declaration signed.	Well organised with all screenshots labelled. One or two tables incomplete, or a minor file-naming error.	Disorganised or several tables blank. Screenshots present but unlabelled or hard to read.	Required structure not followed, screenshots unlabelled or missing, declaration not signed.

PERFORMANCE TASK 3 — SCORE SHEET

To be completed by the instructor.

Criterion	Weight	Score	Comments
Disk Provisioning & Identification	15		
Partitioning Accuracy	25		
Formatting & File Systems	20		
Mounting, File Creation & Clean Unmount	20		
Concept Questions	15		
Documentation Quality	5		
TOTAL	100		

Student: _____ Section: _____ Date: _____

Evaluated by: _____ Signature: _____